

Assignment-2

Akshaya-G

Part-1 A simple binomial model

given (a)

$$f(x; \theta) = \frac{10!}{x!(10-x)!} \theta^x (1-\theta)^{10-x}$$

$$L(\theta|y) = \frac{10!}{y!(10-y)!} \theta^y (1-\theta)^{10-y} \quad (y \text{ fixed})$$

$$P(\theta) = \begin{cases} 1 & \text{when } 0 \leq \theta \leq 1 \\ 0 & \text{when } \theta < 0 \text{ or } \theta > 1 \end{cases}$$

$$P(\theta|y) = \frac{L(\theta|y) P(\theta)}{\int L(\theta|y) P(\theta) d\theta}$$

$$\text{given } y=7, \int L(\theta|y) P(\theta) d\theta = \frac{1}{11}$$

$$\therefore P(\theta|y) = ?$$

$$a) \theta = 0.75$$

$$L(0.75|y) = \frac{\overset{4}{\cancel{10!}} \overset{3}{\cancel{2 \times 7 \times 10}} \theta^7 (1-\theta)^3}{\cancel{7!} \cdot \cancel{3!} \cdot \cancel{7 \times 2}} \theta^7 (1-\theta)^3$$

$$= 120 \times \theta^7 (1-\theta)^3$$

$$= 120 \times (0.75)^7 (0.25)^3$$

$$= 0.2503$$

$$P(\theta|y) = \frac{0.2503}{1/11} = 2.753$$

$$b) \theta = 0.25$$

$$L(0.25|y) = 120 \times (0.25)^7 (0.75)^3$$

$$= 0.00309$$

$$P(\theta|y) = \frac{0.00309}{1/11} = 0.0339$$

① $\theta = 1 \quad L(\theta|y) = 120 \times (1)^7 (0)^3$
 $= 0$

$P(\theta|y) = 0$

1.2. code! graph!

1.3. $P'(\theta) = \frac{d}{d\theta} [1320 \theta^7 (1-\theta)^3]$

$0 = 1320 [7\theta^6 (1-\theta)^3 + (-3(1-\theta)^2 (\theta^7))]$

$7\theta^6 (1-\theta)^2 = 3(1-\theta)^2 (\theta^7)$

$7(1-\theta) = 3\theta$

$7 = 10\theta$

$\theta = \frac{7}{10}$

1.4.

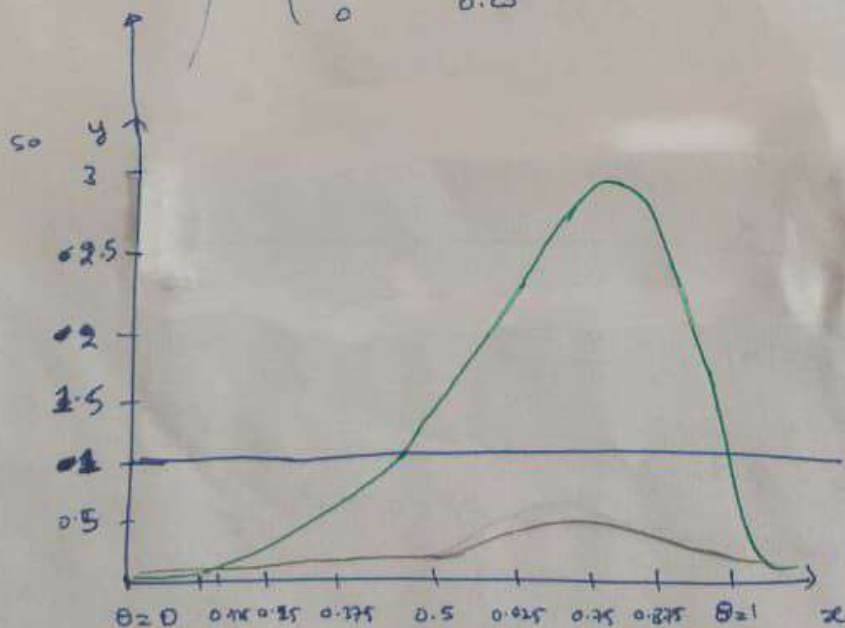
$L(\theta|y) = 120 \theta^7 (1-\theta)^3$

$P(\theta|y) = 1320 \theta^7 (1-\theta)^3$

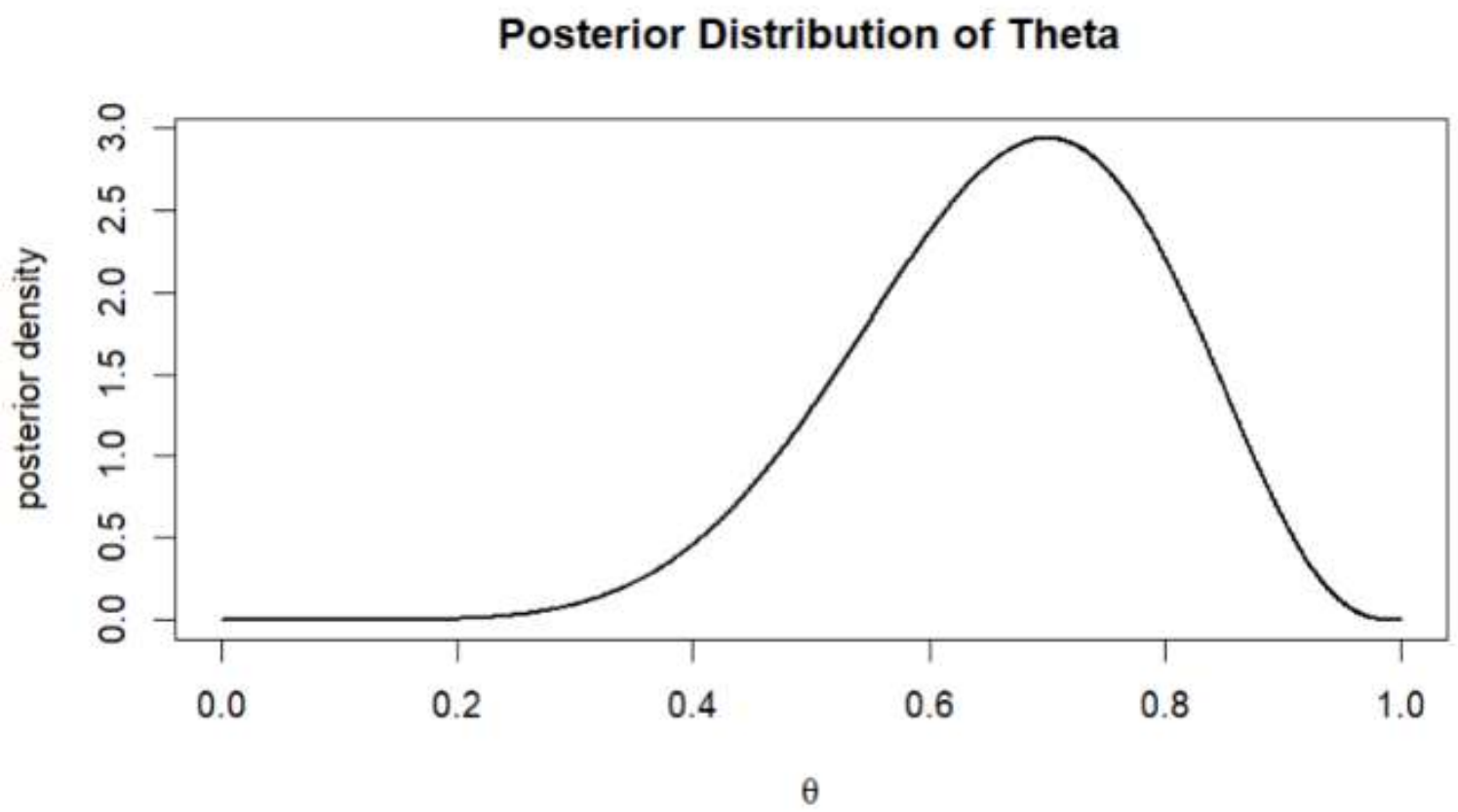
$P(\theta) = \begin{cases} 1 & \text{when } 0 \leq \theta \leq 1 \\ 0 & \text{o.w.} \end{cases}$

compressed version of $P(\theta|y)$
 bell curve.

flat line



question 1.2 graph



question 1.2 code

```
1 # 1.2 Graph the posterior distribution of theta
2 theta <- seq(0, 1, length.out = 1000)
3 posterior_density <- 1320 * (theta^7) * ((1 - theta)^3)
4 plot(theta, posterior_density,
5       type = "l",
6       col = "black",
7       lwd = 2,
8       main = "Posterior Distribution of Theta",
9       xlab = expression(theta),
10      ylab = "posterior density")
11 |
```

11:1 (Top Level) ↕

R Script ↕

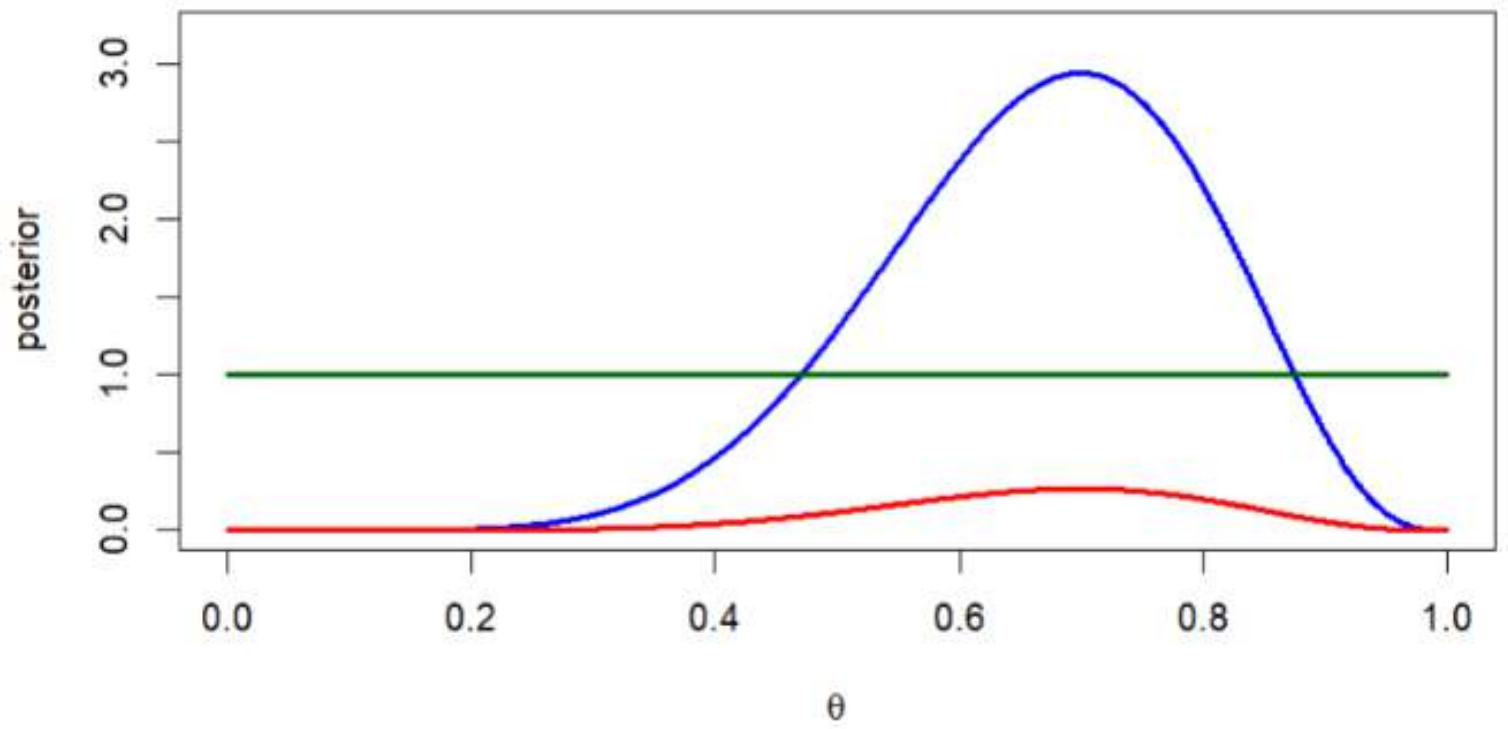
Console Terminal × Background Jobs ×

R • R 4.6.0 • ~/

🔍

```
> # 1.2 Graph the posterior distribution of theta
> theta <- seq(0, 1, length.out = 1000)
> posterior_density <- 1320 * (theta^7) * ((1 - theta)^3)
> plot(theta, posterior_density,
+       type = "l",
+       col = "black",
+       lwd = 2,
+       main = "Posterior Distribution of Theta",
+       xlab = expression(theta),
+       ylab = "posterior density")
> |
```

question 1.4 graph



question 1.4 code

```
1 theta <- seq(0, 1, length.out = 1000)
2 prior <- rep(1, length(theta))
3
4 likelihood <- 120 * (theta^7) * ((1 - theta)^3)
5
6 posterior <- 1320 * (theta^7) * ((1 - theta)^3)
7
8 plot(theta, posterior,
9       type = "l",
10      col = "blue",
11      lwd = 2.5,
12      xlab = expression(theta),
13      ylim = c(0, 3.2))
14
15 lines(theta, likelihood, col = "red", lwd = 2.5)
16
17 lines(theta, prior, col = "darkgreen", lwd = 2.5)
18
19
20
21 |
```

21:1 (Top Level) ↕

R Script ↕

Console Terminal Background Jobs

R 4.6.0 ~ /

```
> likelihood <- 120 * (theta^7) * ((1 - theta)^3)
>
> posterior <- 1320 * (theta^7) * ((1 - theta)^3)
>
> plot(theta, posterior,
+       type = "l",
+       col = "blue",
+       lwd = 2.5,
+       xlab = expression(theta),
+       ylim = c(0, 3.2))
>
>
> lines(theta, likelihood, col = "red", lwd = 2.5)
>
> lines(theta, prior, col = "darkgreen", lwd = 2.5)
>
> |
```

Part 2 - A Gaussian model of reading

data $y \rightarrow n$ i.i.d $y_1, y_2, \dots, y_n$

(a) $y_i \sim \text{Normal}(\mu, \sigma)$

$$L(\mu, \sigma | y) = \frac{1}{(\sigma\sqrt{2\pi})^n} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2}$$

$$\sigma = 50$$

$$\mu \sim \text{Normal}(250, 25)$$

$y \rightarrow n=8 \rightarrow 300, 270, 390, 450, 500, 290, 680, 450$

2.1 $\mu = 300$ $e^{-\frac{1}{2 \times (50)^2} \sum_{i=1}^8 (y_i - \mu)^2}$

$$L(\mu, \sigma | y) = \frac{1}{(50 \times \sqrt{2 \times 3.14})^8} e^{-\frac{1}{2(25)} (\mu - 250)^2}$$

$$P(\mu) = \frac{1}{25\sqrt{2\pi}} e^{-\frac{1}{2(25)} (\mu - 250)^2}$$

$$C = \left(\frac{1}{50\sqrt{2\pi}} \right)^8 \cdot \frac{1}{25\sqrt{2\pi}} \approx 2.621 \times 10^{-10}$$

$$P'(\mu | \sigma, y) = L(\mu, \sigma | y) P(\mu) = C \times e^{-\frac{1}{5000} \sum_{i=1}^8 (y_i - \mu)^2} \times e^{-\frac{(\mu - 250)^2}{1250}}$$

a) $\mu = 300$

$$P'(\mu | \sigma, y) = 2.621 \times 10^{-10} \times e^{-\frac{1}{5000} (0 + 900 + 8100 + 22500 + 40000 + 100 + 14400 + 22500)} \times e^{-\frac{5000}{2 \times 25}}$$

$$= 2.621 \times 10^{-10} \times e^{-47.7-2}$$

$$\approx 7.53 \times 10^{-42}$$

$$\mu = 900$$

$$p'(\mu, y) = c \times e$$

$$\frac{-1}{5000} \left(360000 + 396000 + 260100 + 102500 + 160000 + 372100 + 48400 + 202500 \right) \times e^{\left(\frac{-(900-250)^2}{1250} \right)}$$

$$= 2.621 \times 10^{-20} \times e^{-400.5} \times e^{-338} \approx (3.2 \times 10^{-341})$$

$$\approx 0$$

$$\mu = 50$$

$$p'(\mu, y) = c \times e$$

$$\frac{-1}{5000} \left(62500 + 484000 + 115600 + 160000 + 202500 + 57600 + 396900 + 160000 \right) \times e^{\left(\frac{-(50-250)^2}{1250} \right)}$$

$$= 2.621 \times 10^{-20} \times e^{-(240.7)} \times e^{(-32)} \approx (3.4 \times 10^{-337})$$

$$\approx 0$$

2.3

- 1) The posterior curve shifts significantly to the right because the actual reading times averaged much higher.
- 2) and the unnormalised posterior density values are much much smaller than that of prior

question 2.2 code

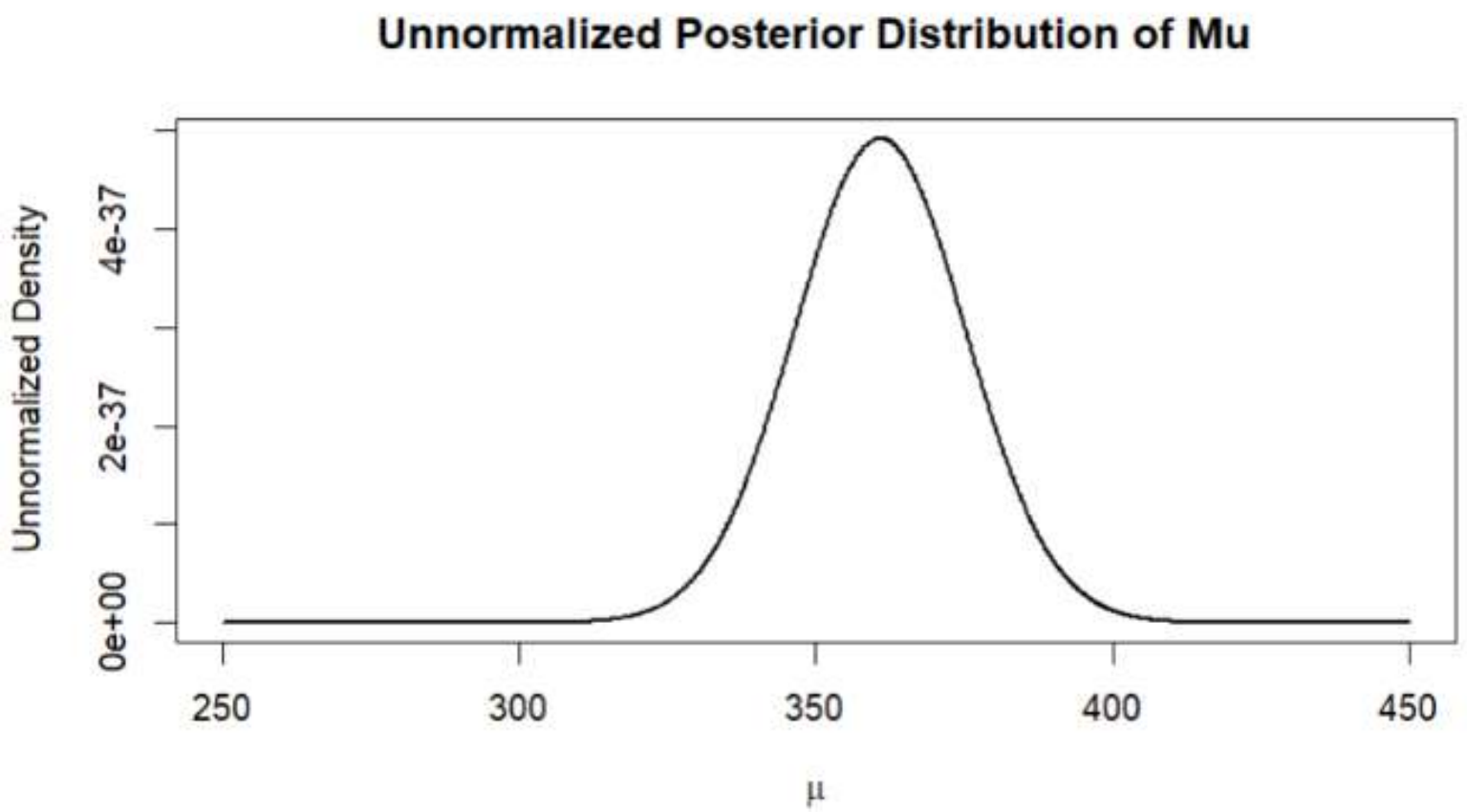
```
1 y <- c(300, 270, 390, 450, 500, 290, 680, 450)
2 sigma <- 50
3 prior_mean <- 250
4 prior_sd <- 25
5
6 mu <- seq(250, 450, length.out = 1000)
7
8 unnormalized_posterior <- sapply(mu, function(m) {
9   likelihood <- prod(dnorm(y, mean = m, sd = sigma))
10  prior <- dnorm(m, mean = prior_mean, sd = prior_sd)
11  return(likelihood * prior)
12 })
13
14 plot(mu, unnormalized_posterior,
15       type = "l",
16       col = "black",
17       lwd = 2,
18       main = "Unnormalized Posterior Distribution of Mu",
19       xlab = expression(mu),
20       ylab = "Unnormalized Density")
21 |
```

21:1 (Top Level) ↕

R Script

```
Console Terminal Background Jobs
R • R 4.6.0 • ~/
> mu <- seq(250, 450, length.out = 1000)
>
> unnormalized_posterior <- sapply(mu, function(m) {
+   likelihood <- prod(dnorm(y, mean = m, sd = sigma))
+   prior <- dnorm(m, mean = prior_mean, sd = prior_sd)
+   return(likelihood * prior)
+ })
>
> plot(mu, unnormalized_posterior,
+       type = "l",
+       col = "black",
+       lwd = 2,
+       main = "Unnormalized Posterior Distribution of Mu",
+       xlab = expression(mu),
+       ylab = "Unnormalized Density")
> |
```

question 2.2 graph



question 2.3 code

```
1 y <- c(300, 270, 390, 450, 500, 290, 680, 450)
2 mu <- seq(150, 500, length.out = 1000)
3
4 prior <- dnorm(mu, mean = 250, sd = 25)
5
6
7 unnormalized_posterior <- sapply(mu, function(m) {
8   likelihood <- prod(dnorm(y, mean = m, sd = 50))
9   prior_val <- dnorm(m, mean = 250, sd = 25)
10  return(likelihood * prior_val)
11 })
12
13 plot(mu, prior,
14       type = "l", col = "red", lwd = 2.5,
15       main = "Graph 1: Prior Distribution",
16       xlab = expression(mu), ylab = "Density")
17
18 plot(mu, unnormalized_posterior,
19       type = "l", col = "blue", lwd = 2.5,
20       main = "Graph 2: Unnormalized Posterior",
21       xlab = expression(mu), ylab = "Unnormalized Density")
22
23
```

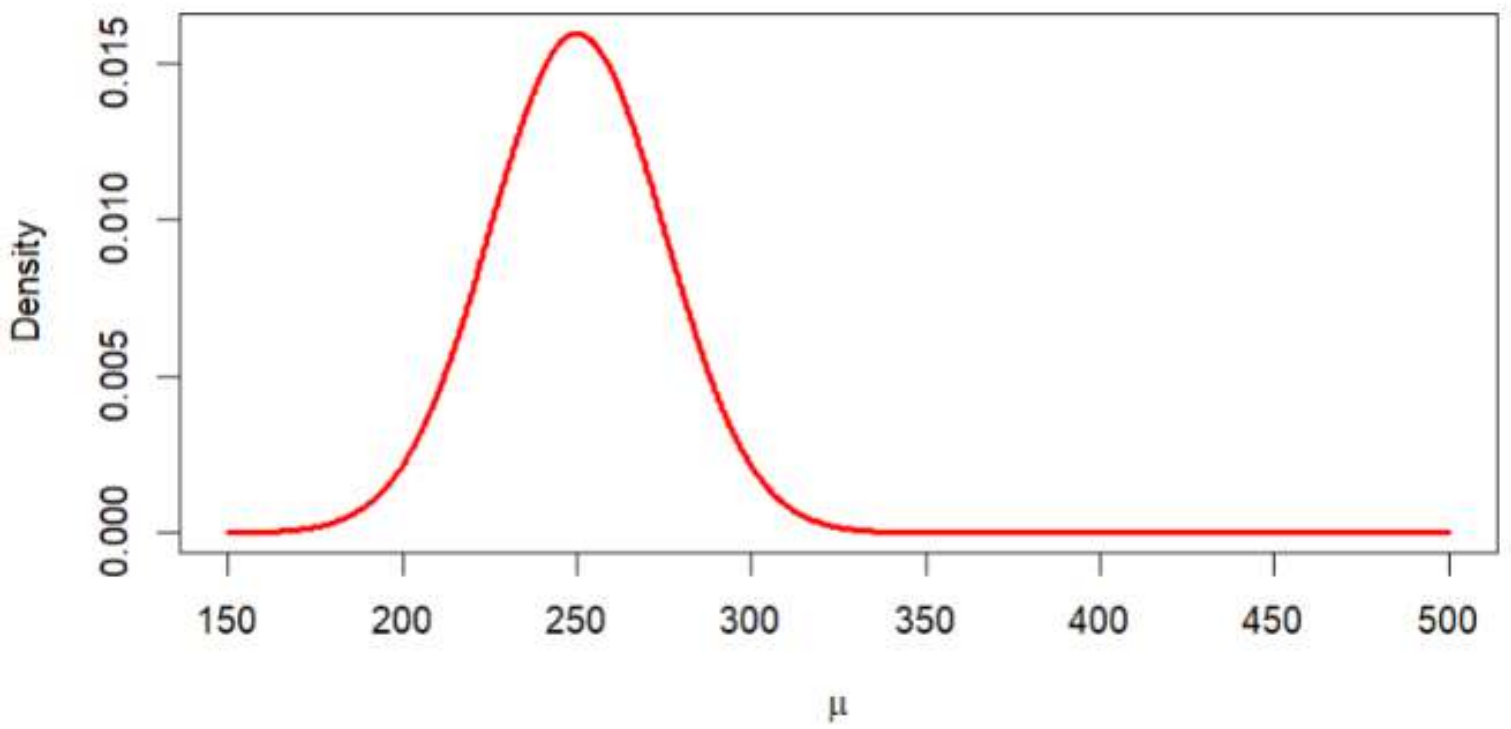
16:46 (Top Level) R Script

Console Terminal Background Jobs

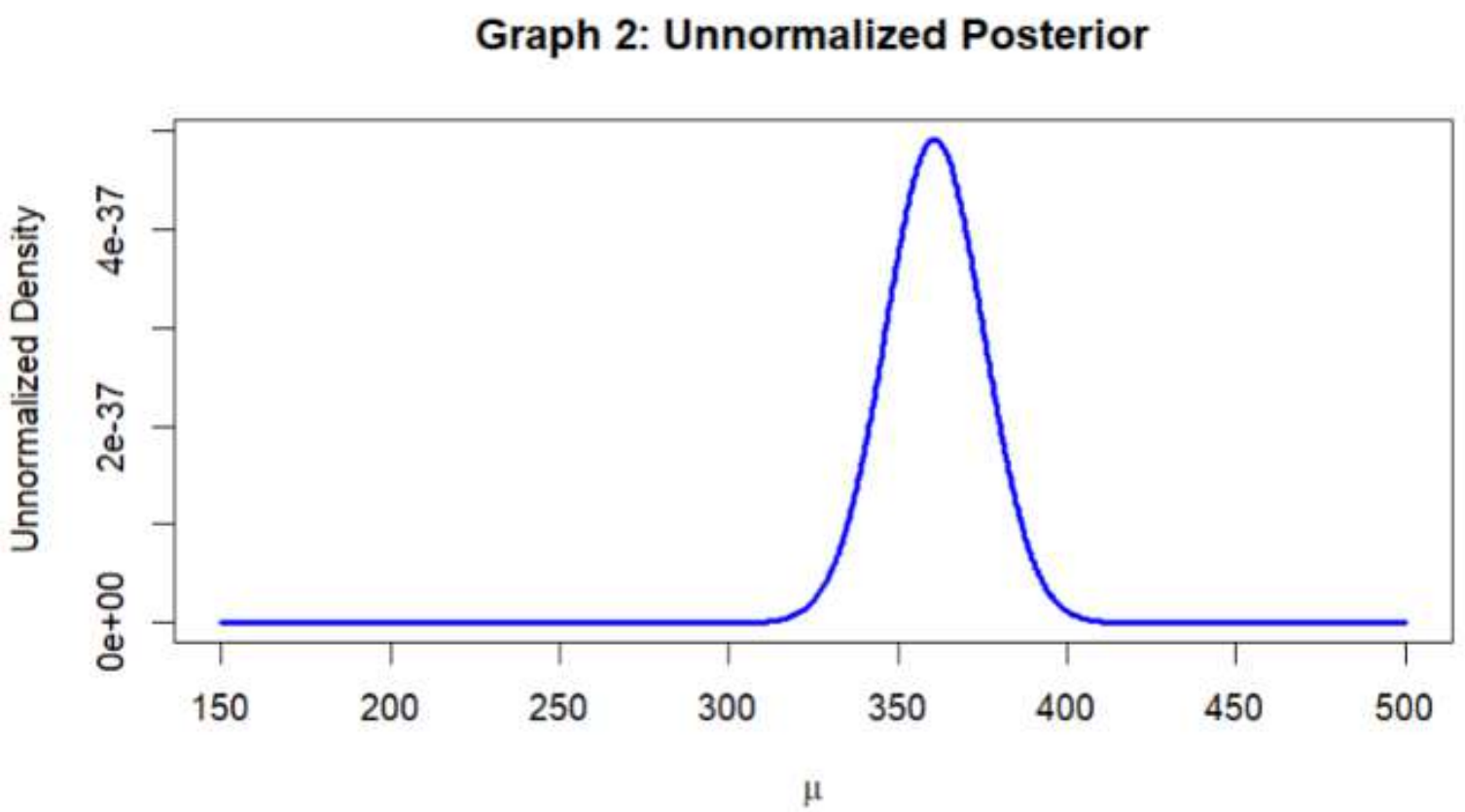
```
> unnormalized_posterior <- sapply(mu, function(m) {
+   likelihood <- prod(dnorm(y, mean = m, sd = 50))
+   prior_val <- dnorm(m, mean = 250, sd = 25)
+   return(likelihood * prior_val)
+ })
>
> plot(mu, prior,
+       type = "l", col = "red", lwd = 2.5,
+       main = "Graph 1: Prior Distribution",
+       xlab = expression(mu), ylab = "Density")
>
> plot(mu, unnormalized_posterior,
+       type = "l", col = "blue", lwd = 2.5,
+       main = "Graph 2: Unnormalized Posterior",
+       xlab = expression(mu), ylab = "Unnormalized Density")
> })
```

question 2.3 graph

Graph 1: Prior Distribution



question 2.3 graph



Part 3 The Bayesian Learning

$$L(d|k) = \frac{d^k e^{-d}}{k!}$$

prior $\rightarrow d \sim \text{Gamma}(40, 2)$

posterior $\rightarrow d \sim \text{Gamma}(40+k, 3)$

$k \rightarrow$ no. of road accidents a day.

data 25, 20, 23, 27.

Day	k	prior	posterior
day 1	25	Gamma(40, 2)	Gamma(65, 3)
day 2	20	Gamma(65, 3)	Gamma(85, 4)
day 3	23	Gamma(85, 4)	Gamma(108, 5)
day 4	27	Gamma(108, 5)	Gamma(135, 6)
day 5	k	Gamma(135, 6)	Gamma(135+k, 6)

3.1 $\rightarrow$ Gamma(135, 6)

3.2 $E[H]$ is the no. of accidents.

$$\begin{aligned}\text{Gamma}(\alpha, \beta) \quad E[H] &= \frac{\alpha}{\beta} \\ &= \frac{135}{6}\end{aligned}$$

$$E[H] = 22.5$$

So the model predicts 22.5 road accidents on day 5.

Part 4: Model building in the Bayesian framework

4.1 $\rightarrow$ n words
 $\rightarrow$ n non words

T_w $\rightarrow$ word recognition time

T_{nw} $\rightarrow$ non-word recognition time

4.2. hypothesis: $\mu_w = \mu_{nw}$ (Null)
 $\mu_w > \mu_{nw}$ (Lexical-access)

4.3 models

	Null	
L	$T_w \sim \text{Normal}(\mu, \sigma)$ $T_{nw} \sim \text{Normal}(\mu + \delta, \sigma)$	$T_w \sim \text{Normal}(\mu, \sigma)$ $T_{nw} \sim \text{Normal}(\mu + \delta, \sigma)$
Prior	$\mu \sim \text{Normal}(300, 50)$ $\sigma = 60$ $\delta = 0$	$\mu \sim \text{Normal}(300, 50)$ $\sigma = 60$ $\delta \sim \text{Normal}_+(0, 50)$

830

Date

	T_w	T_{nw}
1.	285.0780	296.8060
2.	267.5184	280.1157
3.	289.9203	310.4417
4.	379.0674	324.8276
5.	359.9884	373.8154
6.	403.3993	269.8220

4.5 Null $\rightarrow p'(\mu, \delta | T_w, T_{nw}) = L(\mu, \delta, \sigma | T_w) L(\mu, \delta, \sigma | T_{nw}) P(\mu) P(\delta)$

4.5.3 The prior predictions of the Null hypothesis model show completely overlapping distributions for words & non-words because it assumes there is absolutely no time difference ($\delta = 0$). In contrast, the prior predictions for the lexical-access model show the non-word distribution clearly shifted to the right (longer times). This reflects its structural assumption ($\delta > 0$) that processing meaningless non-words inherently takes longer than recognizing familiar words. (graph & Rescorla in upcoming slides)

question 4.5.1 code

```
1
2 dat <- read.table("https://raw.githubusercontent.com/yadavhimanshu059/CGS698C/main/notes/Module-
3 Tw <- dat$Tw
4 Tnw <- dat$Tnw
5
6 sigma <- 60
7 delta <- 0
8
9 mu_grid <- seq(200, 400, length.out = 500)
10
11 unnormalized_posterior_null <- sapply(mu_grid, function(mu) {
12   lik_w <- prod(dnorm(Tw, mean = mu, sd = sigma))
13   lik_nw <- prod(dnorm(Tnw, mean = mu + delta, sd = sigma))
14   prior_mu <- dnorm(mu, mean = 300, sd = 50)
15   return(lik_w * lik_nw * prior_mu)
16 })
17
18 plot(mu_grid, unnormalized_posterior_null,
19       type = "l",
20       col = "blue",
21       lwd = 2.5,
22       main = "4.5.1 Unnormalized Posterior of Mu (Null Model)",
23       xlab = expression(mu),
24       ylab = "Unnormalized Density")
25
```

11:62 <function>(mu) ↕

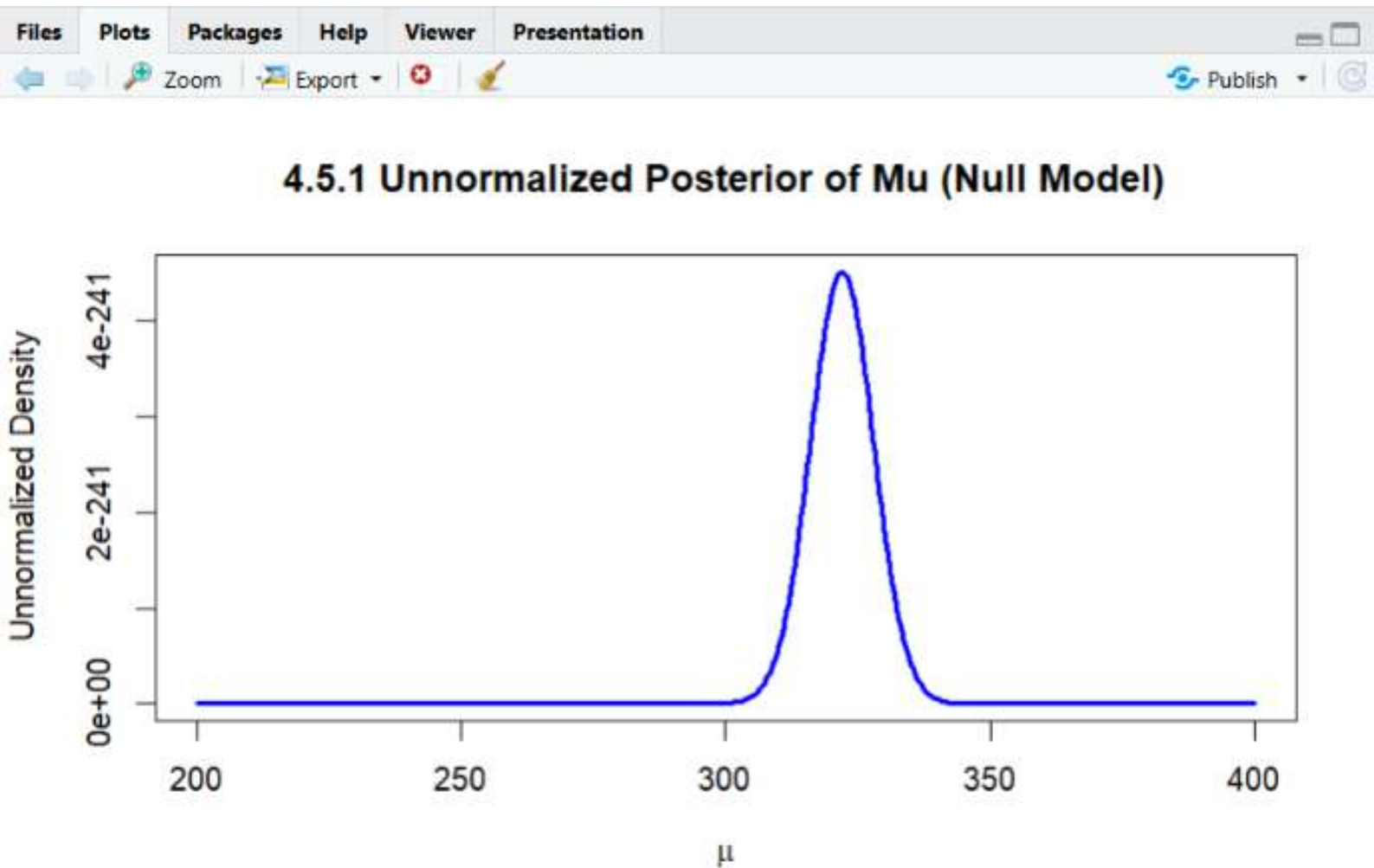
R Script ↕

Console Terminal Background Jobs

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```
+ })
+
+ >
+ > plot(mu_grid, unnormalized_posterior_null,
+       type = "l",
+       col = "blue",
+       lwd = 2.5,
+       main = "4.5.1 Unnormalized Posterior of Mu (Null Model)",
+       xlab = expression(mu),
+       ylab = "Unnormalized Density")
+
+ > |
```

question 4.5.1 graph



question 4.5.2 code

```
1 library(truncnorm)
2
3
4 n_sims <- 10000
5
6 sim_mu <- rnorm(n_sims, mean = 300, sd = 50)
7 sim_delta <- rtruncnorm(n_sims, a = 0, b = Inf, mean = 0, sd = 50)
8
9 sim_Tw <- rnorm(n_sims, mean = sim_mu, sd = 60)
10 sim_Tnw <- rnorm(n_sims, mean = sim_mu + sim_delta, sd = 60)
11
12 par(mfrow = c(1, 2))
13
14 hist(sim_Tw,
15      main = "Prior Predictions: Tw (Words)",
16      xlab = "Recognition Time",
17      col = "lightgreen",
18      breaks = 50)
19
20 hist(sim_Tnw,
21      main = "Prior Predictions: Tnw (Non-words)",
22      xlab = "Recognition Time",
23      col = "lightblue",
24      breaks = 50)
25
26 par(mfrow = c(1, 1))
```

19:1 (Top Level) ⚙

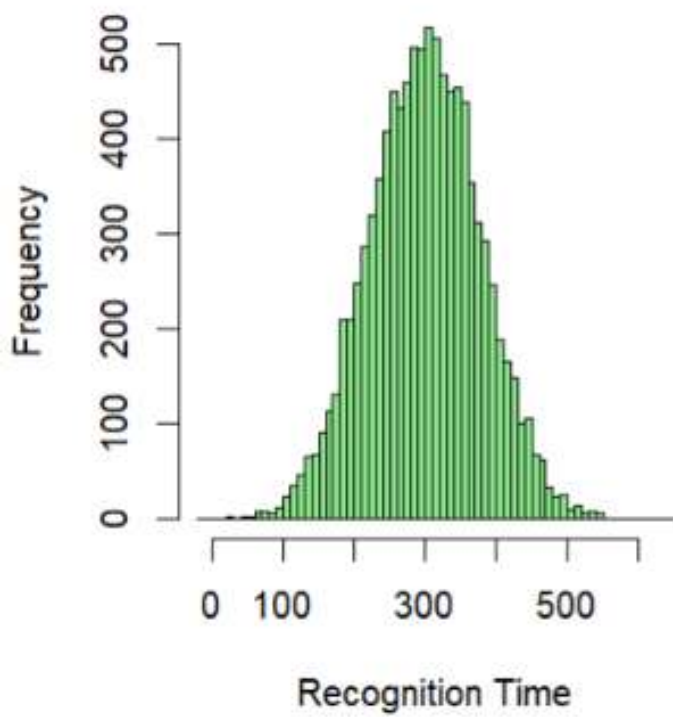
R Script

Console Terminal Background Jobs

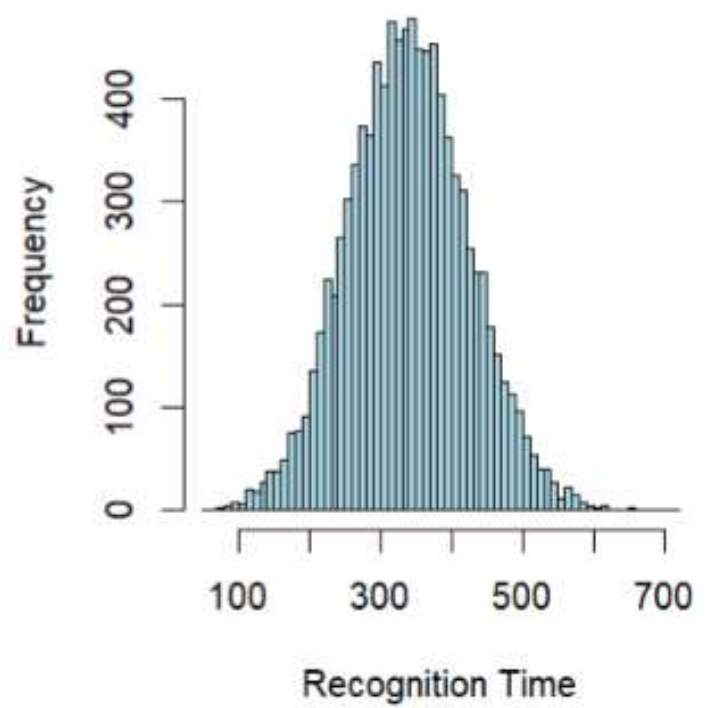
```
R • R 4.6.0 • ~/
+ col = "lightgreen",
+ breaks = 50)
>
> hist(sim_Tnw,
+      main = "Prior Predictions: Tnw (Non-words)",
+      xlab = "Recognition Time",
+      col = "lightblue",
+      breaks = 50)
>
> par(mfrow = c(1, 1))
>
```

question 4.5.2 graph

Prior Predictions: Tw (Words)



Prior Predictions: Tnw (Non-words)



4.5.4. The observed data shows that non-word recognition times (T_{nw}) are clearly shifted to the right (taking longer) compared to word recognition times (T_w). The null hypothesis model incorrectly predicts that both distributions will perfectly overlap, completely missing this shift. Therefore, the lexical-access model is much more consistent with the data, as its prior predictions inherently anticipate the longer processing times for non-words.

question 4.5.4 code

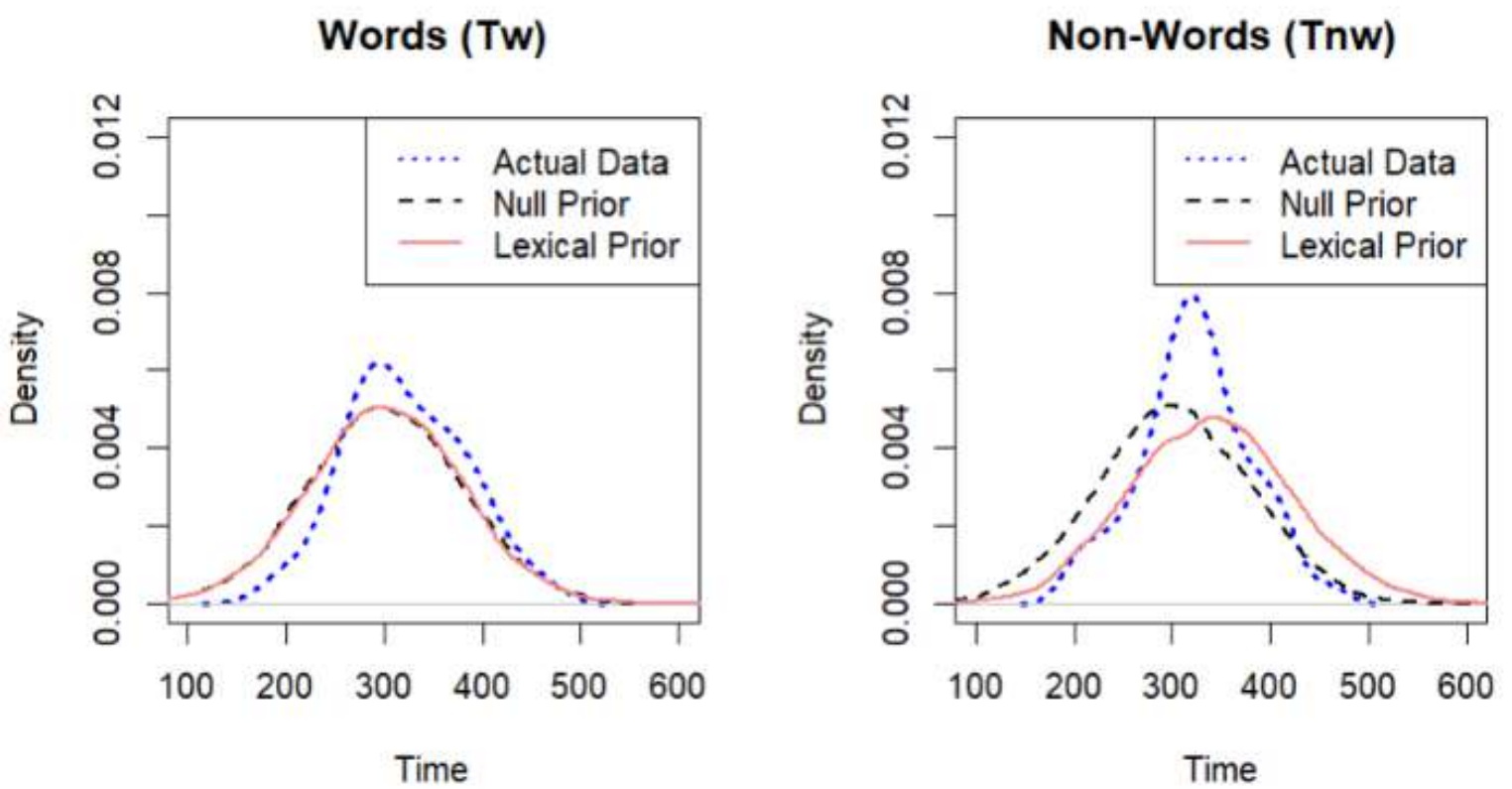
```
1 library(truncnorm)
2
3 dat <- read.table("https://raw.githubusercontent.com/yadavhimanshu059/CGS698C/main/notes/Module-2/recognition.csv", sep=",",
4
5 n <- 10000
6
7 mu_null <- rnorm(n, 300, 50)
8 Tw_null <- rnorm(n, mu_null, 60)
9 Tnw_null <- rnorm(n, mu_null, 60)
10
11 mu_lex <- rnorm(n, 300, 50)
12 delta_lex <- rtruncnorm(n, a=0, b=Inf, mean=0, sd=50)
13 Tw_lex <- rnorm(n, mu_lex, 60)
14 Tnw_lex <- rnorm(n, mu_lex + delta_lex, 60)
15
16 par(mfrow = c(1, 2))
17
18 plot(density(dat$Tw), main="words (Tw)", xlab="Time", col="blue", lwd=3, lty=3, xlim=c(100, 600), ylim=c(0, 0.012))
19 lines(density(Tw_null), col="black", lwd=2, lty=2)
20 lines(density(Tw_lex), col="lightcoral", lwd=2, lty=1)
21 legend("topright", legend=c("Actual Data", "Null Prior", "Lexical Prior"),
22       col=c("blue", "black", "lightcoral"), lty=c(3,2,1), lwd=2)
23
24 plot(density(dat$Tnw), main="Non-Words (Tnw)", xlab="Time", col="blue", lwd=3, lty=3, xlim=c(100, 600), ylim=c(0, 0.012))
25 lines(density(Tnw_null), col="black", lwd=2, lty=2)
26 lines(density(Tnw_lex), col="lightcoral", lwd=2, lty=1)
27 legend("topright", legend=c("Actual Data", "Null Prior", "Lexical Prior"),
28       col=c("blue", "black", "lightcoral"), lty=c(3,2,1), lwd=2)
29
30 par(mfrow = c(1, 1))
```

15:1 (Top Level) R Script

Console Terminal Background Jobs

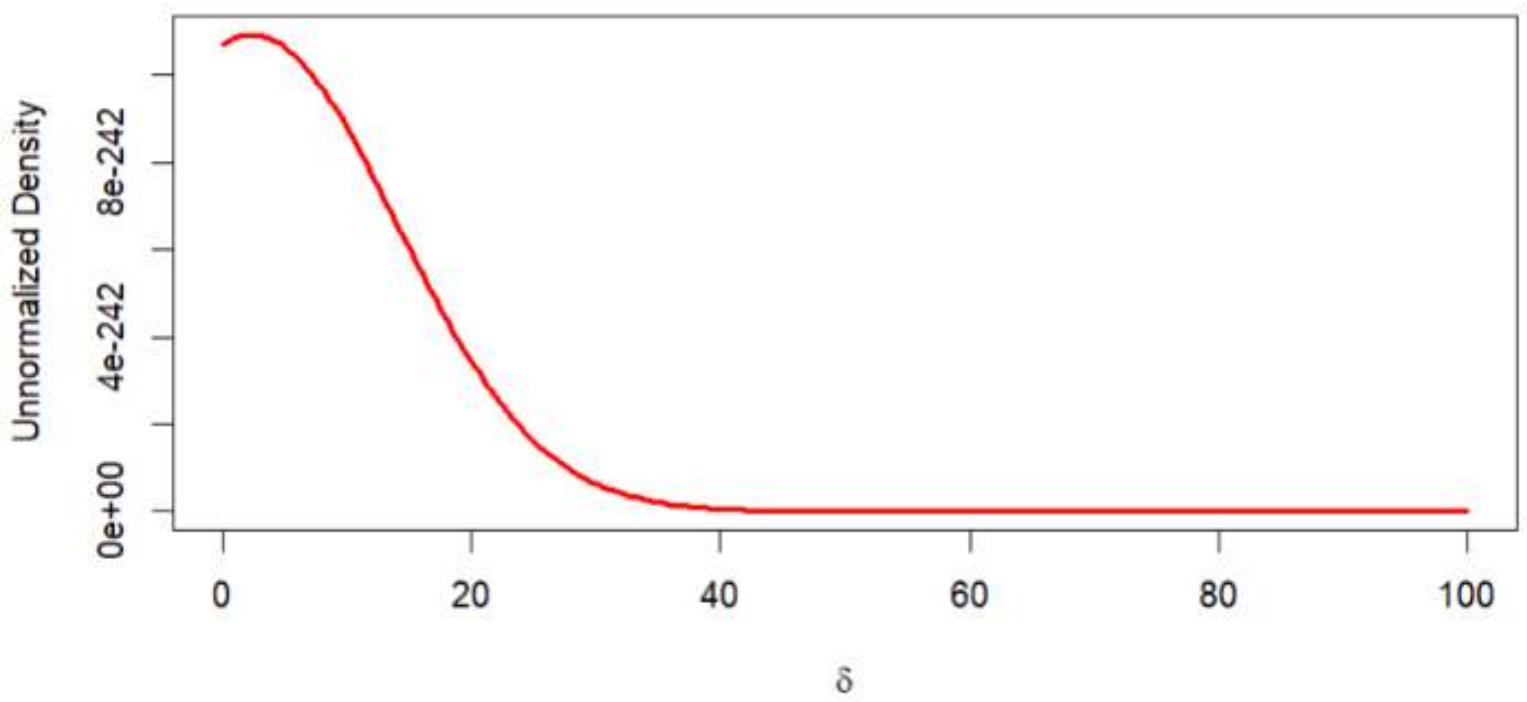
```
R - R 4.6.0 - ~/
> # Graph B: Non-Words (Tnw)
> plot(density(dat$Tnw), main="Non-Words (Tnw)", xlab="Time", col="blue", lwd=3, lty=3, xlim=c(100, 600), ylim=c(0, 0.012))
> lines(density(Tnw_null), col="black", lwd=2, lty=2)
> lines(density(Tnw_lex), col="lightcoral", lwd=2, lty=1)
> legend("topright", legend=c("Actual Data", "Null Prior", "Lexical Prior"),
+       col=c("blue", "black", "lightcoral"), lty=c(3,2,1), lwd=2)
>
```


question 4.5.4 graph



question 4.5.5 graph

4.5.5 Unnormalized Posterior of Delta (Lexical Model)



question 4.5.5 code

```
library(truncnorm)

dat <- read.table("https://raw.githubusercontent.com/yadavhimanshu059/CGS698C/main/notes/Mo
Tw <- dat$Tw
Tnw <- dat$Tnw

sigma <- 60

delta_grid <- seq(0, 100, length.out = 200)
mu_grid <- seq(200, 400, length.out = 200)

unnormalized_posterior_delta <- sapply(delta_grid, function(delta) {

  sum(sapply(mu_grid, function(mu) {
    lik_w <- prod(dnorm(Tw, mean = mu, sd = sigma))
    lik_nw <- prod(dnorm(Tnw, mean = mu + delta, sd = sigma))
    prior_mu <- dnorm(mu, mean = 300, sd = 50)
    prior_delta <- dtruncnorm(delta, a = 0, b = Inf, mean = 0, sd = 50)

    return(lik_w * lik_nw * prior_mu * prior_delta)
  }))
})

plot(delta_grid, unnormalized_posterior_delta,
     type = "l",
     col = "red",
     lwd = 2.5,
     main = "4.5.5 Unnormalized Posterior of Delta (Lexical Model)",
     xlab = expression(delta),
     ylab = "Unnormalized Density")
```

(Top Level) R Script

Terminal × Background Jobs ×

4.6.0 · ~/

```
type = "l",
col = "red",
lwd = 2.5,
main = "4.5.5 Unnormalized Posterior of Delta (Lexical Model)",
xlab = expression(delta),
ylab = "Unnormalized Density")
```

question 4.5.3 code

```
1 library(truncnorm)
2 n_sims <- 10000
3
4 mu_null <- rnorm(n_sims, mean = 300, sd = 50)
5 Tw_null <- rnorm(n_sims, mean = mu_null, sd = 60)
6 Tnw_null <- rnorm(n_sims, mean = mu_null + 0, sd = 60)
7
8 mu_lex <- rnorm(n_sims, mean = 300, sd = 50)
9 delta_lex <- rtruncnorm(n_sims, a = 0, b = Inf, mean = 0, sd = 50)
10 Tw_lex <- rnorm(n_sims, mean = mu_lex, sd = 60)
11 Tnw_lex <- rnorm(n_sims, mean = mu_lex + delta_lex, sd = 60)
12
13 par(mfrow = c(2, 2))
14
15 hist(Tw_null, main="Null Model: Words", xlab="Time", col="lightblue", xlim=c(100,600), breaks=50)
16 hist(Tnw_null, main="Null Model: Non-Words", xlab="Time", col="lightblue", xlim=c(100,600), breaks=50)
17
18 hist(Tw_lex, main="Lexical Model: Words", xlab="Time", col="lightgreen", xlim=c(100,600), breaks=50)
19 hist(Tnw_lex, main="Lexical Model: Non-Words", xlab="Time", col="lightcoral", xlim=c(100,600), breaks=50)
20
21 # Reset layout
22 par(mfrow = c(1, 1))
23 |
```

23:1 (Top Level) ↕

R Script ↕

Console Terminal Background Jobs

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```
> par(mfrow = c(2, 2))
```

```
> hist(Tw_null, main="Null Model: Words", xlab="Time", col="lightblue", xlim=c(100,600), breaks=50)
```

```
> hist(Tnw_null, main="Null Model: Non-Words", xlab="Time", col="lightblue", xlim=c(100,600), breaks=50)
```

```
> hist(Tw_lex, main="Lexical Model: Words", xlab="Time", col="lightgreen", xlim=c(100,600), breaks=50)
```

```
> hist(Tnw_lex, main="Lexical Model: Non-Words", xlab="Time", col="lightcoral", xlim=c(100,600), breaks=50)
```

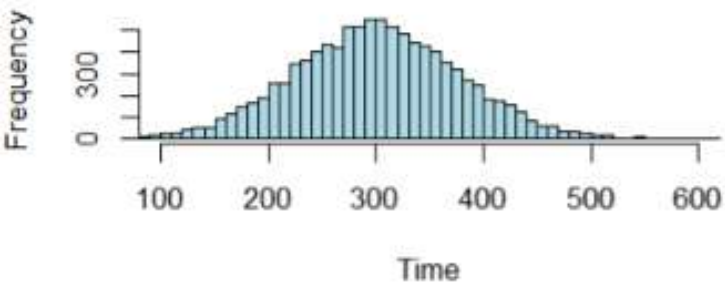
```
> # Reset layout
```

```
> par(mfrow = c(1, 1))
```

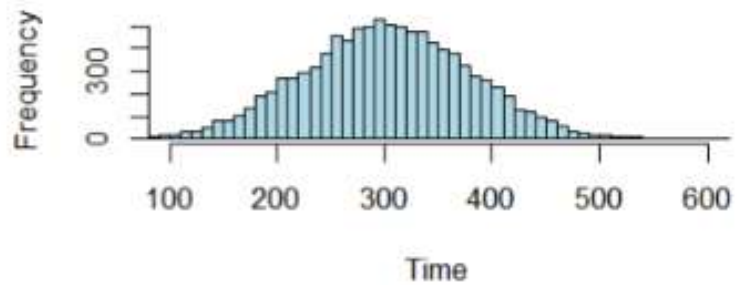
```
>
```


question 4.5.3 graph

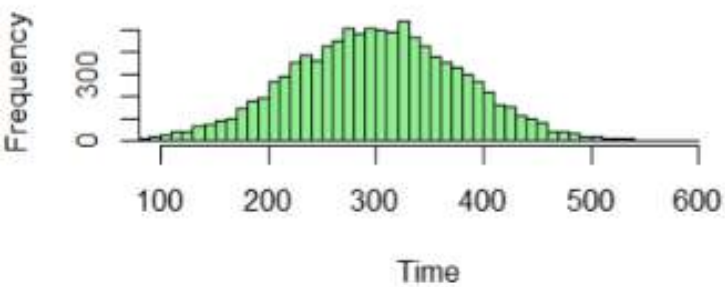
Null Model: Words



Null Model: Non-Words



Lexical Model: Words



Lexical Model: Non-Words

